

===== Newton-Raphson Power Flow Solution – 3 Bus System =====

Y-Bus Matrix (Real part G):

```

20.0000  -10.0000  -10.0000
-10.0000   26.0000  -16.0000
-10.0000  -16.0000   26.0000

```

Y-Bus Matrix (Imaginary part B):

```

-50    20    30
 20   -52    32
 30    32   -62

```

Starting Newton-Raphson Iterations...

Iter	dP2	dQ2	dP3	V2
1	2.86000000	0.22000000	1.43840000	1.00000000
2	0.09921789	0.05091398	0.02171493	0.97345133
3	0.00021661	0.00014300	0.00003819	0.97168415
4	0.00000000	0.00000000	0.00000000	0.97167974

>>> Converged in 4 iterations (tolerance = 1e-06)

===== BUS VOLTAGES (Final Solution) =====

Bus	V (pu)	Angle (deg)	P (pu)	Q (pu)
1	1.050000	0.000000	2.184228	1.408515
2	0.971680	-2.696454	-4.000000	-2.500000
3	1.040000	-0.498803	2.000000	1.461769

===== LINE FLOWS AND LINE LOSSES =====

Line	P_from(pu) P_loss(pu)	Q_from(pu) Q_loss(pu)	P_to(pu)	Q_to(pu)
1-2	1.793618 Loss: 0.083934	1.187339 0.167867	-1.709684	-1.019472
1-3	0.390610 Loss: 0.001828	0.221176 0.005483	-0.388783	-0.215693
2-3	-2.290316 Loss: 0.098467	-1.480528 0.196934	2.388783	1.677462

Total Active Power Loss : 0.184228 pu = 18.4228 MW
Total Reactive Power Loss: 0.370284 pu = 37.0284 Mvar

=====

RESULTS IN PHYSICAL UNITS (100 MVA Base)

Bus	V (pu)	Angle (deg)	P (MW)	Q (Mvar)
1	1.050000	0.000000	218.4228	140.8515
2	0.971680	-2.696454	-400.0000	-250.0000
3	1.040000	-0.498803	200.0000	146.1769

>>